

# Quantitative Research Methods

## Bayesian Inference

Advanced module A12 — optional self-study

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# Where this module fits

| Course chapter / module      | What this module builds on it                                 |
|------------------------------|---------------------------------------------------------------|
| Ch. 0 — statistics refresher | Confidence intervals and $p$ -values — here they get rivals   |
| Ch. 3 — linear regression    | The likelihood; Bayesian regression reuses the whole fit      |
| Ch. 4 — classification       | Bayes' theorem itself (LDA already used it for labels)        |
| Ch. 6 — regularisation       | Ridge <i>is</i> a Bayesian estimate in disguise — proved here |

## The other way to be uncertain

The whole taught course is frequentist: parameters are fixed, data are random. This module flips it — **treat the parameter as the random thing**, update beliefs with data, and read off statements the frequentist machinery cannot make: “the probability that  $p$  exceeds 0.25 is 0.93.”

One of twelve optional advanced modules; self-study. Every number on these slides is reproduced by `make_figures.py` in the module folder.

# Contents

- 1 The Problem
- 2 The Beta-Binomial Engine
- 3 Gaussian Means & Shrinkage
- 4 Computation: MCMC
- 5 Regression & Decisions
- 6 Python Lab
- 7 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

# Notation in this module

| Symbol                                   | Meaning                                                |
|------------------------------------------|--------------------------------------------------------|
| $\theta$                                 | the unknown parameter, now carrying a distribution     |
| $\pi(\theta)$                            | the prior: belief before the data                      |
| $L(\theta) = p(\text{data} \mid \theta)$ | the likelihood (Chapters 3–4's object)                 |
| $\pi(\theta \mid \text{data})$           | the posterior: belief after the data                   |
| $\text{Beta}(a, b)$                      | the workhorse prior for a probability                  |
| CrI                                      | credible interval (posterior probability interval)     |
| MAP                                      | maximum a posteriori: the posterior's peak             |
| $\tau$                                   | prior standard deviation (how sure you are beforehand) |

## Sources and references

### Primary textbook

James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning, with Applications in Python*, Springer. — Supplies the likelihoods, ridge, and the datasets (Advertising.csv).

### Canonical further reading (optional)

Gelman et al., *Bayesian Data Analysis* (3rd ed., 2013); McElreath, *Statistical Rethinking* (2nd ed., 2020) — the standard references, one formal, one narrative.

# Learning objectives

By the end of this module you will be able to:

- **State** Bayes' theorem for parameters and the roles of prior, likelihood and posterior.
- **Update** a Beta prior with binomial data by hand, and report a posterior mean and credible interval.
- **Explain** every posterior mean in this module as a *precision-weighted compromise* between prior and data.
- **Derive** ridge regression as the MAP estimate under a Gaussian prior — and translate  $\lambda \leftrightarrow \tau$ .
- **Implement** a Metropolis sampler from scratch and check it against the analytic answer.
- **Distinguish** credible from confidence intervals without garbling either.

# Roadmap of this module

## Believe, observe, update

- 1 The problem: what kind of probability a parameter can have.
- 2 The Beta-binomial engine: updating by hand, priors, credible intervals.
- 3 Gaussian means and shrinkage: precision weighting — and ridge unmasked.
- 4 Computation: when conjugacy runs out, walk (Metropolis).
- 5 Bayesian regression and decisions: Advertising, and an A/B call.
- 6 Python lab, summary, appendix.

# Outline

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## Two questions that sound the same

| Frequentist (Ch. 0)                                           | Bayesian (this module)                                              |
|---------------------------------------------------------------|---------------------------------------------------------------------|
| How would my estimate vary across imaginary repeated samples? | What should I believe about $\theta$ , given the one sample I have? |
| Parameter fixed, data random                                  | Data fixed, parameter random                                        |
| $p$ -value, confidence interval                               | posterior, credible interval                                        |
| "95% of such intervals cover"                                 | "95% probability $\theta$ is in here"                               |

### Rule (Bayes' theorem for parameters)

$$\pi(\theta \mid \text{data}) \propto L(\theta) \pi(\theta)$$

$$\text{posterior} \propto \text{likelihood} \times \text{prior.}$$

### How to read this

Chapter 4's LDA already ran Bayes' theorem — on *class labels*. The only new move is to run it on *parameters*. The likelihood is the same object Chapters 3–4 maximised; the prior is new; the posterior is the answer.

## Exercise A12.1 — Credible or confident? [Concept]

### Exercise

A 95% *confidence* interval for a conversion rate is  $[0.19, 0.31]$ ; a 95% *credible* interval from the same data is  $[0.19, 0.31]$  as well.

- 1 Write the one-sentence claim each interval is entitled to make.
- 2 A manager says: “there is a 95% chance the true rate is between 0.19 and 0.31.” Which interval licenses that sentence?
- 3 Why do the two intervals often nearly coincide anyway?

## Solution — Exercise A12.1

### Solution

**Part 1.** Confidence: “this interval was built by a recipe that captures the true rate in 95% of repeated samples” — a statement about the *procedure*. Credible: “given the data and the prior, the posterior probability that the rate lies in  $[0.19, 0.31]$  is 0.95” — a statement about *this* interval.

**Part 2.** Only the credible interval. The manager’s sentence assigns a probability to the parameter, which frequentist logic forbids: once computed, a confidence interval either contains the fixed truth or does not. (Everyone reads confidence intervals the manager’s way — Chapter 0 flagged the misreading; here it becomes a licensed statement.)

**Part 3.** With a flat-ish prior and decent  $n$ , the posterior is essentially the normalised likelihood, so both intervals track the same curvature — they agree numerically while meaning different things. The divergence appears when priors are informative or  $n$  is small.

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# Conjugacy: the update you can do in your head

## Rule (Beta-binomial conjugacy)

prior  $\text{Beta}(a, b)$  +  $y$  successes in  $n$  trials  $\implies$  posterior  $\text{Beta}(a + y, b + n - y)$

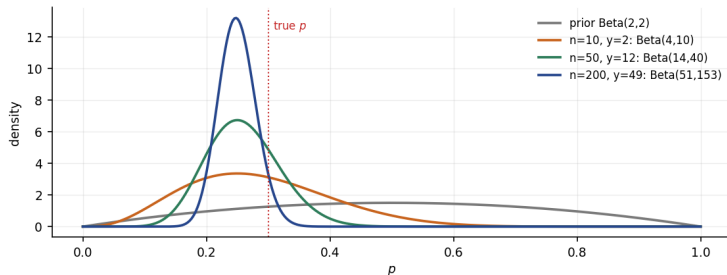
## How to read this

The prior acts like  $a + b$  *imaginary trials*,  $a$  of them successes. Updating is bookkeeping: add real successes to  $a$ , real failures to  $b$ . Posterior mean  $= \frac{a+y}{a+b+n}$  — always *between* the prior mean and the sample proportion.

## A conversion rate, true $p = 0.3$

Prior  $\text{Beta}(2, 2)$ . After  $n = 10$  ( $y = 2$ ):  $\text{Beta}(4, 10)$ , mean 0.286. After  $n = 50$  ( $y = 12$ ):  $\text{Beta}(14, 40)$ , mean 0.259. After  $n = 200$  ( $y = 49$ ):  $\text{Beta}(51, 153)$ , mean 0.250, 95% CrI [0.193, 0.312].

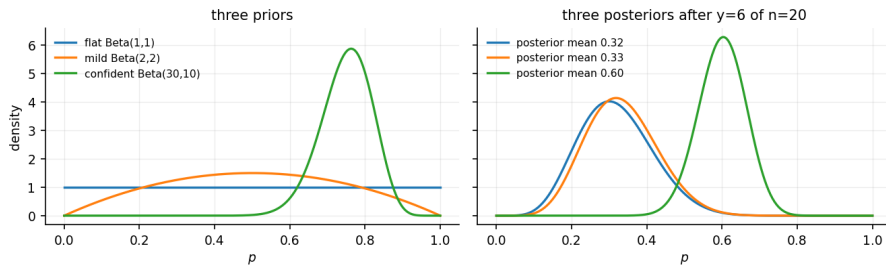
# Watching belief sharpen



## Data overwhelm the prior — at the data's own pace

The gentle Beta(2, 2) hump narrows steadily; by  $n = 200$  the posterior is tight around the sample rate and the prior's four imaginary trials are outvoted 50 : 1. The truth (0.3) sits inside every credible interval along the way.

# Three priors, one dataset



## Priors matter exactly as much as they claim to

Same data ( $y = 6$  of  $n = 20$ ). Flat prior  $\rightarrow$  mean 0.318; mild Beta(2, 2)  $\rightarrow$  0.333; a confident-but-wrong Beta(30, 10) (forty imaginary trials at 75%)  $\rightarrow$  0.600 — the data are outvoted 2:1. A prior is a claim with a stake; make it inspectable, and show the sensitivity.

## Exercise A12.2 — One update by hand [Math]

### Exercise

Prior  $\text{Beta}(2, 2)$  on a click-through rate; you observe 7 clicks in 10 impressions.

- 1 Write down the posterior distribution.
- 2 Compute the prior mean, the MLE, and the posterior mean.
- 3 Show that the posterior mean is a weighted average  $w \cdot \text{MLE} + (1 - w) \cdot \text{prior mean}$ , and identify  $w$ .



## Solution — Exercise A12.2

### Solution

**Part 1.**  $\text{Beta}(2 + 7, 2 + 3) = \text{Beta}(9, 5)$ .

**Part 2.** Prior mean =  $2/4 = 0.5$ ; MLE =  $7/10 = 0.7$ ; posterior mean =  $9/14 \approx 0.643$ .

**Part 3.** With  $a + b = 4$  prior trials and  $n = 10$  real ones,

$$\frac{a + y}{a + b + n} = \underbrace{\frac{n}{n + a + b}}_{w=10/14} \cdot \frac{y}{n} + \underbrace{\frac{a + b}{n + a + b}}_{1-w=4/14} \cdot \frac{a}{a + b} = \frac{10}{14}(0.7) + \frac{4}{14}(0.5) = 0.643.$$

The weight on the data is *sample size against imaginary sample size* — the entire prior-vs-data politics of this module in one fraction.

### Common mistake

Reporting the MAP ( $8/12 = 0.667$ ) as “the Bayesian estimate” without saying so. Mean, median and MAP all summarise the posterior; for skewed posteriors they differ, and the honest report is the whole interval.

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# The Gaussian mean: precision-weighted compromise

Rule (known  $\sigma$ , prior  $\mu \sim N(\mu_0, \tau^2)$ )

$$\text{posterior mean} = w \bar{x} + (1 - w) \mu_0, \quad w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2}$$

posterior variance =  $(n/\sigma^2 + 1/\tau^2)^{-1}$ .

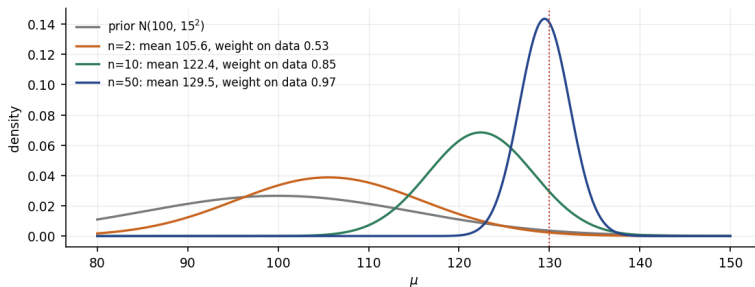
## How to read this

Precisions (inverse variances) *add*; means combine *weighted by precision*. The data's vote grows linearly in  $n$ ; the prior's vote is fixed — the Beta-binomial story again, in Gaussian dress.

**Prior  $N(100, 15^2)$ ,  $\sigma = 20$ , truth 130**

$n = 2$ : mean 105.6, weight on data 0.53.  $n = 10$ : 122.4, weight 0.85.  $n = 50$ : 129.5, weight 0.97 — the prior fades on schedule.

# The prior fading, drawn



## Shrinkage is not a bug

At  $n = 2$  the posterior barely moves off the prior — and that is the *correct* behaviour: two noisy observations should not overturn a considered belief. Every shrinkage estimator in Chapter 6 is this picture wearing an optimisation costume, as the next slide proves.

# Ridge regression, unmasked

## Claim

Ridge (Ch. 6) is the **MAP estimate** of a linear model under the prior  $\beta_j \sim N(0, \tau^2)$ , with  $\lambda = \sigma^2 / \tau^2$ .

## The two-line proof

Maximising the log-posterior = log-likelihood + log-prior:

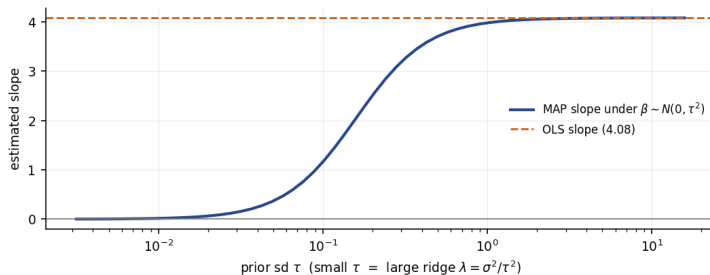
$$-\frac{1}{2\sigma^2} \sum_i (y_i - x_i^\top \beta)^2 - \frac{1}{2\tau^2} \sum_j \beta_j^2 \propto -\left[ \sum_i (y_i - x_i^\top \beta)^2 + \frac{\sigma^2}{\tau^2} \sum_j \beta_j^2 \right],$$

which is exactly the ridge criterion with  $\lambda = \sigma^2 / \tau^2$ .

## Chapter 6's tuning dial, translated

A large penalty  $\lambda$  is a tight prior ( $\tau$  small). Cross-validating  $\lambda$  empirically chooses how strong a prior the data can support. (The lasso? A Laplace prior — same construction.)

# The dictionary, drawn on Advertising



## Read the dial

Standardised sales-on-TV slope: OLS says 4.08. As the prior tightens ( $\tau \downarrow$ ), the MAP slope shrinks toward 0 — at  $\tau = 0.1$  (i.e.  $\lambda = 500$ ) it is 1.16. Chapter 6's whole coefficient-path figure is this curve, one predictor at a time.

## Exercise A12.3 — Precision weighting by hand [Math]

### Exercise

Quality control believes a machine fills bottles at  $\mu_0 = 100\text{ml}$  ( $\tau = 15$ ). Measurement noise is  $\sigma = 20$ . A morning sample gives  $n = 4$ ,  $\bar{x} = 120$ .

- 1 Compute the posterior mean and standard deviation of  $\mu$ .
- 2 The engineer says “the sample says 120, so recalibrate to 120.” What does the posterior say instead, and why?
- 3 How large would  $n$  (at  $\bar{x} = 120$ ) have to be for the weight on the data to reach 0.9?

## Solution — Exercise A12.3

### Solution

**Part 1.** Precisions: prior  $1/\tau^2 = 1/225 \approx 0.00444$ ; data  $n/\sigma^2 = 4/400 = 0.01$ . Weight on data  $w = 0.01/0.01444 \approx 0.692$ . Posterior mean  $= 0.692 \times 120 + 0.308 \times 100 \approx 113.8$ ; posterior sd  $= \sqrt{1/0.01444} \approx 8.3$ .

**Part 2.** Split the difference toward 113.8: with only four noisy measurements ( $SE = 20/2 = 10$ ), a 20ml jump from a well-calibrated prior is as likely to be sampling noise as drift. The posterior *is* the rational compromise — and its sd (8.3) says a follow-up sample is worth taking before touching the machine.

**Part 3.** Need  $\frac{n/400}{n/400 + 1/225} = 0.9 \Rightarrow n/400 = 9/225 \Rightarrow n = 16$ .

### Common mistake

Adding standard deviations instead of precisions. Only *inverse variances* add; the posterior sd is always *smaller* than both the prior's and the data's alone.



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## When conjugacy runs out, walk

Real models rarely have closed-form posteriors. The general tool: build a random walk whose long-run visiting frequencies *are* the posterior.

### Algorithm (Metropolis)

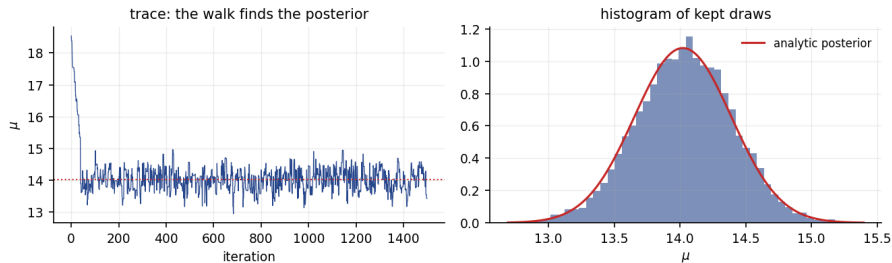
From current  $\theta$ : (1) propose  $\theta' = \theta + N(0, s^2)$ ; (2) accept with probability  $\min(1, \pi(\theta' | \text{data})/\pi(\theta | \text{data}))$ , else stay; (3) record, repeat. Discard the warm-up (“burn-in”), keep the rest.

### Why it works, in one sentence

Moves uphill are always taken and moves downhill sometimes — calibrated so the walk spends time in each region *in proportion to its posterior probability*; only posterior *ratios* are needed, so the intractable normalising constant cancels.

*Like A2's Shapley values and A3's conformal intervals, we build it in 15 lines; the production tools are PyMC, Stan, NumPyro.*

# Metropolis on real data: the mean of Advertising sales



## Sampler vs analytic truth

20,000 steps, acceptance rate 0.62, burn-in 2,000. Posterior mean 14.04 vs analytic 14.02; posterior sd 0.367 vs analytic 0.369. The trace (left) starts at a deliberately wrong value and finds the posterior within a few hundred steps.

## Exercise A12.4 — Reading a sampler [Concept]

### Exercise

Three colleagues run Metropolis on the same posterior and show you their diagnostics:

- 1 Colleague A: acceptance rate 0.99; the trace drifts in slow waves.
- 2 Colleague B: acceptance rate 0.02; the trace is long flat plateaus.
- 3 Colleague C: acceptance rate 0.55; the trace is a fuzzy caterpillar.

Diagnose each, name the shared root cause for A and B, and state the fix.

## Solution — Exercise A12.4

### Solution

**A — steps too small.** Nearly every proposal is accepted because it barely moves; the chain explores the posterior like a snail, and consecutive draws are almost copies. Estimates converge, but painfully slowly.

**B — steps too large.** Proposals leap into low-probability territory and are almost always rejected; the chain freezes for hundreds of iterations. Same disease as A — *high autocorrelation in the chain* — reached from the opposite direction.

**C — healthy.** Mid-range acceptance with fast mixing; rules of thumb put the sweet spot around 0.2–0.5 for random-walk Metropolis (our run: 0.62, acceptably brisk for one dimension).

**Fix.** Tune the proposal scale  $s$  toward the sweet spot; then confirm with the trace and by running a second chain from a different start and checking both agree.

### Common mistake

Judging a sampler by acceptance rate alone. It is a symptom, not the disease: the disease is autocorrelation, and the trace plot (plus multiple chains) is the diagnostic that sees it.

## Extended Exercise A12.1 — Your own sampler [Python]

### Extended exercise (15 min)

Implement Metropolis from scratch for the mean of Advertising sales (treat  $\sigma$  as the sample sd; prior  $\mu \sim N(0, 100^2)$ ):

- 1 Write the log-posterior and the accept/reject loop; run 20,000 steps at proposal scale 0.5 from the (bad) start  $\bar{x} + 5$ .
- 2 Report acceptance rate, posterior mean and sd after burn-in; compare with the analytic values (14.02, 0.369).
- 3 Re-run with proposal scales 0.01 and 50. Connect what you see to Exercise A12.4.

## Solution — Extended Exercise A12.1 (1/2) — code

```

import numpy as np, pandas as pd
x = pd.read_csv("Advertising.csv", index_col=0)["sales"].to_numpy()
rng = np.random.default_rng(2024)
sigma, tau, B, s = x.std(ddof=1), 100.0, 20_000, 0.5

def logpost(mu): # log posterior, up to a constant
    return -0.5*((x - mu)**2).sum()/sigma**2 - 0.5*mu**2/tau**2

chain, cur = np.empty(B), x.mean() + 5
lp = logpost(cur); acc = 0
for i in range(B):
    prop = cur + rng.normal(0, s)
    lp_prop = logpost(prop)
    if np.log(rng.uniform()) < lp_prop - lp: # uphill: always; downhill: sometimes
        cur, lp, acc = prop, lp_prop, acc + 1
    chain[i] = cur
kept = chain[2000:]
print(f"accept {acc/B:.2f}, mean {kept.mean():.2f}, sd {kept.std(ddof=1):.3f}")

```

## Solution — Extended Exercise A12.1 (2/2)

### Solution — parts 2 and 3

**Part 2.** Acceptance  $\approx 0.62$ ; posterior mean 14.04, sd 0.367 — against analytic 14.02 and 0.369. A dozen lines of NumPy reproduce the exact answer to two decimals; that agreement *is* the test that the sampler is correct.

**Part 3.** At  $s = 0.01$  the acceptance rate rockets toward 1 and the trace crawls (colleague A); at  $s = 50$  acceptance collapses toward 0 and the trace plateaus (colleague B). Both chains still *eventually* target the right posterior — the theory guarantees the destination, tuning buys the arrival time.

### Common mistake

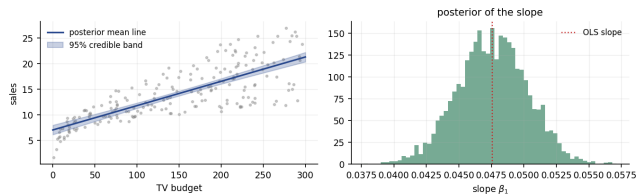
Using  $\pi(\theta')/\pi(\theta)$  on the raw (non-log) scale. With  $n = 200$  the likelihood underflows `float64` long before the ratio is formed — always accept/reject on the *log* scale, as the code does.



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# Bayesian simple regression: sales on TV



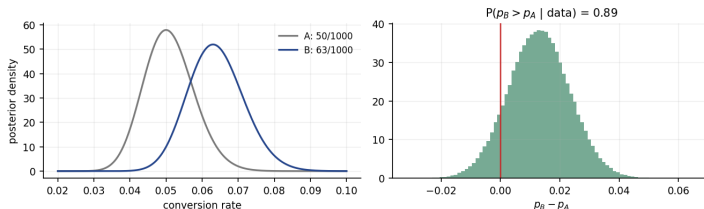
## The posterior of a Chapter 3 slope

Weak  $N(0, 10^2)$  priors on both coefficients. Posterior slope: mean 0.0476, sd 0.0027 — numerically the OLS fit (0.0475) with its standard error, *reinterpreted*: the left panel's band is a 95% credible band of regression lines drawn from the posterior.

## Weak priors reproduce Chapter 3 — by design

With flat-ish priors, Bayesian regression rediscovers least squares; the machinery pays when priors carry information — small  $n$ , correlated predictors (ridge!), hierarchies of related groups.

# A decision, not a $p$ -value: the Bayesian A/B test



A: 50/1000, B: 63/1000

Flat priors, two independent Beta posteriors, 200,000 paired draws:  $\Pr(p_B > p_A | \text{data}) = 0.895$  — a directly decision-shaped number. (The two-sided frequentist test at these counts does not clear 0.05; both summaries are honest, they answer different questions.)

## Common mistake

Believing posteriors are immune to module A1's peeking problem: if you *stop when the posterior looks good*, the long-run error rate of that *policy* still inflates — design the stopping rule, whatever your school.

## Extended Exercise A12.2 — The Bayesian A/B memo [Integrative]

### Extended exercise (15 min)

Marketing ran the experiment above (A: 50/1000, B: 63/1000) and asks for a recommendation memo.

- 1 Reproduce  $\Pr(p_B > p_A \mid \text{data})$  by simulation with flat priors. State the number with its meaning in one sentence.
- 2 Compute the posterior of the *lift*  $p_B - p_A$  and report a 95% credible interval. Does it contain 0, and does that contradict part 1?
- 3 Switching costs EUR 2,000; each conversion is worth EUR 5 on 100,000 annual visitors. Turn the posterior into an expected-value decision.
- 4 State one sentence of prior sensitivity: what would a sceptical prior (e.g. Beta(50, 950) on both arms) do to the conclusion?

## Solution — Extended Exercise A12.2

### Solution

**Part 1.** Draw from  $\text{Beta}(51, 951)$  and  $\text{Beta}(64, 938)$ , compare:  $\Pr(p_B > p_A) \approx 0.895$  — “given the data, B is better with probability about 0.9.”

**Part 2.** The lift’s posterior is centred near 0.013 with 95% CrI roughly  $[-0.007, 0.033]$  — it *does* straddle 0. No contradiction: the CrI asks “what values are plausible”, part 1 asks “which arm is more likely better”; a 0.895 direction is compatible with an interval touching zero.

**Part 3.** Expected lift  $\approx 0.013 \times 100,000 \times \text{EUR } 5 = \text{EUR } 6,500/\text{year}$  against EUR 2,000 once; even weighting the downside draws, expected value favours switching — recommend B, note the  $\sim 10\%$  chance it is not better, and (module A1) confirm with a pre-registered follow-up if the cost were larger.

**Part 4.** A sceptical  $\text{Beta}(50, 950)$  prior (five percent, held with 1000 trials’ conviction) roughly halves the estimated lift and pulls  $\Pr(p_B > p_A)$  toward 0.8 — the direction survives, the enthusiasm moderates. Report both runs.

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# Python lab — run it live

## Reproduce every number: `make_figures.py`

Every number on these slides is computed by `make_figures.py` in this module's folder, from the bundled `Advertising.csv` and seeded simulations — run it and read it alongside the deck:

- the Beta-binomial updates ( $\text{Beta}(2, 2) \rightarrow \text{Beta}(51, 153)$ ) and the three-priors picture;
- the precision-weighted Gaussian posteriors ( $w = 0.53 \rightarrow 0.97$ );
- the from-scratch Metropolis run (14.04 vs analytic 14.02);
- Bayesian regression on `Advertising` and the ridge-MAP dial;
- the A/B posterior ( $\Pr(p_B > p_A) = 0.895$ ).

A companion notebook in the course lab format is in preparation; until it lands, the script *is* the lab — it runs top to bottom in the course environment with no extra packages.

# The whole module in twelve lines

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(2024)

# belief -> data -> updated belief, all in one conjugate line
a, b = 2, 2 # prior Beta(2,2): four imaginary trials
y, n = 49, 200 # the data
post = stats.beta(a + y, b + n - y)
print(f"posterior mean {post.mean():.3f}, "
      f"95% CrI [{post.ppf(0.025):.3f}, {post.ppf(0.975):.3f}]")

# and the sentence the frequentist interval cannot say:
print(f"P(p > 0.25 | data) = {1 - post.cdf(0.25):.2f}")
```



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# Module A12 in one slide

## What we accomplished

A second, coherent grammar for uncertainty — run on the course's own likelihoods and datasets.

- Posterior  $\propto$  likelihood  $\times$  prior — Chapter 4's theorem, aimed at parameters.
- Beta-binomial updating by hand: Beta(2, 2) plus 200 trials  $\rightarrow$  Beta(51, 153), CrI [0.193, 0.312].
- Every posterior mean is a precision-weighted compromise; the data's weight grows as  $n/(n + \text{imaginary } n)$ .
- Ridge unmasked: MAP under  $N(0, \tau^2)$  priors,  $\lambda = \sigma^2/\tau^2$  — Chapter 6 was Bayesian all along.
- Metropolis from scratch:  $14.04 \pm 0.367$  against analytic  $14.02 \pm 0.369$ .
- Decision-shaped output:  $\Pr(p_B > p_A \mid \text{data}) = 0.895$  — with the peeking caveat imported from A1.

# Key formulas at a glance

**Bayes for parameters**  $\pi(\theta \mid \text{data}) \propto L(\theta) \pi(\theta)$ .

**Beta-binomial**  $\text{Beta}(a, b) + (y, n) \rightarrow \text{Beta}(a+y, b+n-y)$ ; mean  $(a+y)/(a+b+n)$ .

**Shrinkage weight** posterior mean =  $w \hat{\theta}_{\text{data}} + (1-w) \theta_{\text{prior}}$ ,  $w = n/(n+a+b)$  (Beta) or  $\frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2}$  (Gaussian).

**Gaussian posterior** precision =  $n/\sigma^2 + 1/\tau^2$ ; precisions add, never sds.

**Ridge as MAP**  $\hat{\beta}_{\text{ridge}} = \arg \max$  posterior under  $\beta_j \sim N(0, \tau^2)$ , with  $\lambda = \sigma^2/\tau^2$ .

**Metropolis** accept  $\theta'$  w.p.  $\min\{1, \pi(\theta' \mid \text{data})/\pi(\theta \mid \text{data})\}$  — ratios only, log scale in code.

# Vocabulary checklist

| Term              | Meaning                                                     |
|-------------------|-------------------------------------------------------------|
| prior             | belief about $\theta$ before the data, as a distribution    |
| likelihood        | $p(\text{data} \mid \theta)$ — Chapters 3–4's fit criterion |
| posterior         | updated belief; the module's universal output               |
| conjugate prior   | family the posterior stays inside (Beta for binomial)       |
| credible interval | interval holding stated posterior probability               |
| MAP               | posterior mode; ridge and lasso are MAPs                    |
| shrinkage         | posterior pulled from data toward prior                     |
| burn-in           | discarded warm-up of an MCMC chain                          |
| acceptance rate   | share of Metropolis proposals taken; tune toward mid-range  |
| prior sensitivity | re-running with other priors; always reported               |

## Decision rules of thumb

- Small  $n$ , real prior knowledge, or a decision to make  $\Rightarrow$  Bayesian machinery earns its keep. Large  $n$ , flat priors  $\Rightarrow$  it will agree with Chapter 0; use whichever is cheaper.
- Choose priors you can defend out loud, in units of imaginary observations — and always show a sensitivity run.
- Report the whole posterior (or its CrI), not just a point; say *which* point if you must give one.
- Conjugate first, grid second, MCMC when you must; always check a sampler against a case with a known answer.
- Trace plot + second chain before trusting any MCMC number.
- Probability statements about parameters: credible intervals only — never launder a confidence interval into one.

# Common pitfalls

## Don't

- 1 Read a confidence interval as “95% probability the parameter is inside” — that sentence needs a posterior.
- 2 Hide an informative prior (Beta(30, 10) moved the answer from 0.32 to 0.60 on the same data).
- 3 Add standard deviations where only precisions add.
- 4 Compute likelihood ratios off the log scale ( $n = 200$  already underflows).
- 5 Trust a chain by its acceptance rate alone — plateaus and snails both lie.
- 6 Peek-and-stop on a posterior and claim the error rate of a fixed- $n$  design.

# Self-check questions

## Quick self-assessment

- ❶ Prior  $\text{Beta}(2, 2)$ , then 7 successes in 10 trials: posterior, posterior mean, and the weight on the data?
- ❷ Prior  $N(100, 15^2)$ ,  $\sigma = 20$ ,  $n = 4$ ,  $\bar{x} = 120$ : posterior mean?
- ❸ State the ridge–Bayes dictionary in one equation.
- ❹ Your Metropolis run has acceptance 0.99 and a slowly drifting trace. Diagnosis and fix?
- ❺  $\Pr(p_B > p_A \mid \text{data}) = 0.895$  while the 95% CrI for  $p_B - p_A$  includes 0. Contradiction?
- ❻ What is the one sentence a credible interval may say that a confidence interval may not?

## Answers

(1)  $\text{Beta}(9, 5)$ ;  $9/14 \approx 0.643$ ;  $w = 10/14 \approx 0.71$ . (2)  $w \approx 0.69$ , mean  $\approx 113.8$ . (3)  $\lambda = \sigma^2/\tau^2$  under  $\beta_j \sim N(0, \tau^2)$ . (4) Proposal scale too small — enlarge  $s$  toward acceptance  $\sim 0.2$ – $0.5$ , confirm on the trace. (5) No: one is a statement about direction, the other about plausible magnitudes; both are read off the same posterior. (6) “Given the data, the parameter lies in this interval with probability 0.95.”

# Five things to remember

## Module A12 in five lines

- 1 **Posterior  $\propto$  likelihood  $\times$  prior** — one line, the whole framework.
- 2 **Every estimate is a compromise weighted by precision** — the data's weight is  $n$  against the prior's imaginary  $n$ .
- 3 **Ridge is a prior in disguise:  $\lambda = \sigma^2/\tau^2$**  — Chapter 6 has been Bayesian all along.
- 4 **MCMC trades formulas for a walk** — and a from-scratch Metropolis matched the analytic answer to two decimals.
- 5 **Priors are claims with stakes**: state them, price them in imaginary trials, and show the sensitivity run.



# What's next

- **Module A1 — Randomised Controlled Trials:** the A/B decision slide is A1's experiment wearing posterior clothes — and A1's peeking warning survives the change of school.
- **Chapter 6, re-read:** the ridge and lasso paths are prior strength dialled from  $\infty$  to 0; the Bayesian reading explains *why* shrinking helps when  $p$  is large.
- **Module A3 — Conformal Prediction:** the frequentist route to calibrated uncertainty — guarantees without priors, at the price of answering only about *predictions*. Reading A3 and A12 together maps the whole uncertainty landscape.

# Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

## Contents of the appendix

- **The Beta-binomial update, derived** — two lines of algebra; moved out because the recipe, not the proof, carries the deck.
- **A conjugate-pairs reference table** — look-up material.
- **The posterior predictive** — forecasting a new observation, not a parameter; the bridge to Chapters 3 and A3.

# The Beta–binomial update, derived

Prior density  $\pi(p) \propto p^{a-1}(1-p)^{b-1}$ ; binomial likelihood  $L(p) \propto p^y(1-p)^{n-y}$ . Multiply:

$$\pi(p | y) \propto p^y(1-p)^{n-y} \cdot p^{a-1}(1-p)^{b-1} = p^{(a+y)-1}(1-p)^{(b+n-y)-1},$$

which is the kernel of  $\text{Beta}(a+y, b+n-y)$  — no integral was ever computed; recognising the *shape* suffices.

## Why conjugacy is a big deal

The prior and posterior live in the same family, so updating is arithmetic on two counters  $(a, b)$  — streaming data update beliefs in  $O(1)$ , and the “imaginary trials” reading of the prior is literal.

# Conjugate pairs at a glance

| Likelihood                       | Conjugate prior    | Posterior update               |
|----------------------------------|--------------------|--------------------------------|
| Binomial ( $n, p$ )              | Beta( $a, b$ )     | $a+y, b+n-y$                   |
| Poisson ( $\lambda$ )            | Gamma( $a, b$ )    | $a + \sum y_i, b + n$          |
| Gaussian mean (known $\sigma$ )  | $N(\mu_0, \tau^2)$ | precision-weighted (main deck) |
| Gaussian variance (known $\mu$ ) | Inverse-Gamma      | shape + $n/2$                  |
| Multinomial                      | Dirichlet          | add the category counts        |

## How to use this table

Each row is the same story: the prior is phrased as pseudo-data, and updating adds real data to it. Module A4's exponential-family machinery is the deep reason this table exists: every exponential-family likelihood has a conjugate family of exactly this form.

# The posterior predictive: betting on the next observation

For a new observation  $\tilde{y}$ , integrate predictions over the posterior:

$$p(\tilde{y} \mid \text{data}) = \int p(\tilde{y} \mid \theta) \pi(\theta \mid \text{data}) d\theta.$$

## The next conversion

After Beta(51, 153):  $\Pr(\tilde{y} = 1 \mid \text{data}) = E[p \mid \text{data}] = 51/204 = 0.25$  — but for the next *twenty* visitors the predictive is Beta-binomial, *wider* than Binomial(20, 0.25), because parameter uncertainty rides along.

## The bridge to prediction intervals

Chapter 3's prediction intervals and module A3's conformal sets solve this same problem in frequentist dress: honest forecasts must carry *both* noise and parameter uncertainty. The posterior predictive is the Bayesian statement of that requirement.